

A Detection-Bias-Corrected Temporal Event-Stream Dataset for Dynamic Plant–Pollinator Link Prediction

Abstract

We present a reprocessed, machine-learning-ready version of a public H.J. Andrews LTER plant–pollinator visitation dataset (2011–2018, 2021; 180 plots, 18 meadows, 147 plant species, 800 pollinator taxa), purpose-built for dynamic link prediction with Temporal Graph Neural Networks. Starting from two raw survey files (74,942 visitation rows; 23,642 flowering-phenology rows), we separate genuine interactions from structural non-detections. We then correct for imperfect pollinator detection using a Royle-type Poisson N -mixture model, fitted through vectorized numerical marginal-likelihood maximization. Next, we compute a phenology-based confidence weight for each interaction. Finally, we assemble the data into a continuous-time weighted event stream and apply a leakage-safe chronological split. The resulting dataset comprises 20,107 weighted interaction events and 1,389 empirically grounded negative (surveyed, non-detected) events. We document every processing decision, including two pitfalls specific to these data: a silent date-format mismatch between the source files and a real multi-year survey gap that can lead to incorrect results with naive calendar-based splitting. These details are provided to ensure that the dataset can be reused reliably.

Keywords: Temporal graph learning, dynamic link prediction, plant–pollinator networks, N -mixture model, detection bias, ecological network data

Specifications Table

Table 1 lists the details of the raw input datasets.

Value of the Data

- This is, to our knowledge, the first release of the H.J. Andrews plant–pollinator interaction data as a continuous-time weighted event stream suitable for direct use with TGNN / dynamic link prediction frameworks (e.g., TGN, TGAT, and related architectures), rather than as a static interaction matrix.
- The dataset separates true interaction strength from detection artifacts via an explicit statistical model, rather than treating raw visit counts as ground truth; this is directly useful to ecologists studying network robustness, competition, and niche overlap, as well as to ML researchers benchmarking noise-robust temporal graph methods.
- The dataset includes empirically observed negative examples (surveyed sessions with no detected interaction), which are more informative than the randomly corrupted negatives typically used to evaluate link prediction, thereby improving the realism of benchmark evaluations.
- The chronological train/validation/test split, along with the documented multi-year survey gap it accounts for, protects downstream users from a subtle and easy-to-miss data-leakage issue.
- The processing pipeline is fully scripted and reusable; it can be re-executed on future releases of the same LTER survey or adapted to structurally similar plant–pollinator, host–parasite, or other bipartite ecological interaction datasets.

Table 1: Specification table

Subject	Computer Science — Machine Learning (Graph Representation Learning); Ecology — Network Ecology
Specific subject area	Temporal graph learning and dynamic link prediction; construction of a detection-bias-corrected, weighted temporal event stream from a public plant–pollinator visitation survey, for benchmarking Temporal Graph Neural Networks (TGNNs)
Type of data	Tables (CSV); Figure
Data collection	Field observations were collected by the H.J. Andrews Experimental Forest LTER program via standardized timed pollinator watches and flowering-phenology surveys at fixed meadow plots (2011–2018, 2021; no surveys in 2019–2020). This dataset reprocesses those records: harmonizing source files, separating genuine interactions from structural non-detections, correcting detection bias via a hierarchical Poisson N -mixture model, computing phenology-based confidence weights, and assembling a chronological event stream with train/val/test splits.
Data source location	H.J. Andrews Experimental Forest and adjoining Willamette National Forest meadows, Cascade Range, Oregon, USA (approx. 44.20–44.28° N, 122.13–122.15° W); 180 plots across 18 meadows in 5 meadow complexes
Data accessibility	Original raw data: Plant Pollinator data at H.J. Andrews Experimental Forest, 2011 to 2021. Full data page: https://andlter.forestry.oregonstate.edu/data/abstract.aspx?dbcode=SA026

Table 2: Files comprising the dataset.

File	Rows $\times$ Cols	Description
SA02601_v6.csv (raw input)	74,942 $\times$ 26	Raw visitation events: one row per observed pollinator visit at a given plot, date, watch session, and minute; includes species codes, visit count (NO_INT), and a PPI_STATUS flag distinguishing real interactions from surveyed-but-empty watches.
SA02602_v5.csv (raw input)	23,642 $\times$ 16	Raw flowering phenology: one row per plant species surveyed at a given plot, date, and watch session; includes stalk count (NO_STALK) and flower count (NO_FLWS).
plant_pollinator_events.csv	20,107 $\times$ 12	Processed weighted positive event stream: $(\tau, \text{plant_id}, \text{pollinator_id}, c)$ plus intermediate confidence components and provenance columns.
plant_pollinator_negative_events.csv	1,389 $\times$ 5	Structural negatives: watch sessions in which the plot was surveyed but no interaction was observed (PPI_STATUS = 0 in the raw data).
events_train/val/test.csv	16,085 / 2,011 / 2,011 rows	Chronological 80/10/10 split of the positive event stream by cumulative elapsed time, for leakage-safe model training and evaluation.

1. Data Description

The dataset comprises two raw source files (retained for provenance) and five processed files derived from them; Table 2 summarizes each file and Table 3 documents every column of the primary processed file.

Dataset-level summary statistics (computed directly from the processed files): 20,107 weighted positive events span 3,662 days (≈ 10.0 years) across 4,532 distinct plant–pollinator pairs, out of $147 \times 800 = 117,600$ possible pairs — an observed-pair density of 3.85%. This is consistent with the 3.5% density reported for an earlier static-network use of the same underlying survey data by Fu et al. [2], serving as an external validation that our reprocessing has preserved the network’s intrinsic sparsity structure. The final edge weight c has a median of 0.096 (IQR 0.046–0.193); its two components, `confidence_posterior` and `confidence_flowering`, have medians of 0.34 and 0.32, respectively (both IQRs spanning roughly 0.16–0.5), indicating that neither component dominates the product. Only 0.24% of events required the flowering-confidence fallback described in Section 2.

2. Experimental Design, Materials and Methods

2.1. Raw data and known structural properties

The two source files use different date formats (SA02601: DD-MM-YYYY; SA02602: YYYY-MM-DD) — parsing both with a single format string silently produces incorrect or null dates for one file. Therefore, each file is parsed separately and validated by asserting that the parsed year matches the recorded YEAR column, as implemented in Algorithm 1. Furthermore, the source data contain a multi-year sampling gap: no surveys were conducted in 2019 or 2020 (Figure 1a). Any downstream splitting or temporal-decay modeling that assumes uniform sampling must account for this gap explicitly.

2.2. Separating interactions from structural negatives

Raw visitation rows carry a PPI_STATUS flag: PPI_STATUS = 1 denotes a row with a recorded plant–pollinator interaction (61,025 of 74,942 rows); PPI_STATUS = 0 denotes a watch session in which the plot was surveyed but no interaction was observed (13,917 rows, with species-code fields empty by construction). We retain the latter as a separate table of structural negatives rather than discarding them, since they constitute empirically observed absences rather than merely missing data; Algorithm 2 details this separation.

2.3. Detection-bias correction via a Poisson N -mixture model

Raw visit counts underestimate true interaction frequency because pollinators are imperfectly detected during a timed watch. We model this explicitly using a Royle-type N -mixture formulation [4]. For a group $g = (\text{plot}, \text{year}, \text{plant}, \text{pollinator})$ with replicate watches $k = 1, \dots, K_g$ (every WATCH conducted at that plot in that year; an unrecorded pair on a conducted watch counts as an implicit zero):

$$N_g \sim \text{Poisson}(\lambda_g) \quad (1)$$

$$Y_{g,k} | N_g \sim \text{Binomial}(N_g, p) \quad (2)$$

where N_g is the latent true interaction intensity and $Y_{g,k}$ is the detected count on replicate watch k .

The latent abundance N_g is marginalized out numerically — for a truncated range of candidate N (from $\max(y_g)$ up to $\max(y_g) + 30$), we compute

$$P(y_g | \lambda, p) = \sum_N \text{Poisson}(N; \lambda) \prod_{k=1}^{K_g} \text{Binomial}(y_{g,k}; N, p), \quad (3)$$

Table 3: Column-level description of `plant_pollinator_events.csv`, the primary processed output.

Column	Description
<code>tau</code>	Elapsed time since the first record in the dataset, in days (integer, range 0–3,662). Used as the continuous temporal axis for TGNN training; not the raw <code>WATCH</code> ordinal, which would misrepresent the multi-year gap between 2018 and 2021.
<code>date</code>	Calendar date of the observation (ISO 8601), retained for interpretability and for reconstructing calendar-year subsets.
<code>plant_id</code> / <code>pollinator_id</code>	Canonical species codes (union vocabulary across both raw files; 147 plant and 800 pollinator identifiers).
<code>NO_INT</code>	Raw visit count summed within the (plot, year, watch, plant, pollinator) cell (i.e., across duplicate within-watch minute-level rows).
<code>confidence_posterior</code>	Detection-bias-corrected interaction confidence in $[0, 1]$: normalized $\log(1 + \hat{\lambda})$, where $\hat{\lambda}$ is the posterior mean of the latent interaction intensity from the fitted N -mixture model (see Section 2).
<code>confidence_flowering</code>	Phenology-based confidence in $[0, 1]$: log-transformed, within-species-normalized flower count at the time of observation.
<code>c</code>	Final edge weight, $c = \text{confidence_posterior} \times \text{confidence_flowering}$, used as the event weight in the temporal graph.
<code>is_imputed_flowering</code>	Boolean flag; <code>True</code> for the 0.24% of events where no same-session flowering record existed and a last-observation-carried-forward value was substituted (see Section 2).
<code>PLOT_ID</code> / <code>YEAR</code> / <code>WATCH</code>	Provenance fields retained from the raw data, enabling traceability back to the original survey record and supporting alternative (e.g., per-plot or per-meadow) splitting strategies.

fully vectorized over a grid of λ via NumPy broadcasting (Listing 1) rather than looping over candidate λ values. This vectorization enables fitting all 17,472 groups in seconds, avoiding the prohibitive runtime of a naive Python-level loop. Before evaluating this likelihood, each group must be assembled into its replicate-watch count vector y_g and replicate count K_g ; Algorithm 3 performs this grouping using the full set of watches conducted at each plot-year (from *vis*) so that unobserved pairs on conducted watches are recorded as implicit zeros rather than omitted.

A single pooled detection probability p is identified by grid search over informative groups (those with ≥ 2 replicate watches conducted at that plot in that year, i.e., $K_g \geq 2$; 17,342 of the 17,472 total groups qualify, from which 250 are sampled for the pooled fit), profiling out each group’s own λ ; the fitted value on this dataset is $\hat{p} = 0.15$. Given $\hat{p}$, a grid-based Bayesian posterior over λ_g (flat prior) is computed for every group, and its posterior mean $\hat{\lambda}_g$ is retained. Note that $K_g \geq 2$ counts repli-

cate opportunities (watches conducted at that plot-year), not watches on which the pair was actually detected — a stricter criterion requiring ≥ 2 actual detections is satisfied by only 2,092 groups, but is not what the pooling step conditions on.

As an explicit methodological note: the natural presence probability, $P(N_g > 0) = 1 - \exp(-\hat{\lambda}_g)$, saturates almost immediately — its median across all 17,472 groups is 0.985 (Figure 1b, orange). Once a single detection is observed at $\hat{p} = 0.15$, the model is highly confident that a true interaction exists. Although statistically sound, this causes the presence probability to saturate near 1 for nearly every observed pair, rendering it ineffective as a graded edge weight. We instead export the posterior mean λ_g and derive

$$\text{confidence_posterior} = \text{minmax}\left(\log(1 + \hat{\lambda}_g)\right) \quad (4)$$

normalized globally (Figure 1b, blue), which preserves relative interaction intensity.

Algorithm 1 HarmonizeRawData — load and validate the two source files

Require: *vis_path*, *flw_path*

Ensure: validated tables *vis*, *flw* with parsed date columns

```

1: vis  $\leftarrow$  ReadCSV(vis_path)
2: flw  $\leftarrow$  ReadCSV(flw_path)
3: vis.date  $\leftarrow$  ParseDate(vis.SAMPLEDATE, "DD-MM-YYYY")
4: flw.date  $\leftarrow$  ParseDate(flw.SAMPLEDATE, "YYYY-MM-DD")
5: assert vis.date.year = vis.YEAR for every row
6: assert flw.date.year = flw.YEAR for every row
7: return vis, flw

```

Algorithm 2 PartitionInteractions — separate E^+ from E^0

Require: harmonized visitation table vis

Ensure: positive interactions E^+ , structural negatives E^0

```

1: assert  $vis.PPI\_STATUS \in \{0, 1\}$  for every row
2:  $E^+ \leftarrow \{r \in vis : r.PPI\_STATUS = 1\}$ 
3:  $E^0 \leftarrow \{r \in vis : r.PPI\_STATUS = 0\}$ 
4: assert  $E^+.PLTSP\_CODE$  is non-null for every row;  $E^0.PLTSP\_CODE$  is null for every row
5:  $E^+.NO\_INT \leftarrow \text{FillNA}(E^+.NO\_INT, 1)$  ▷ treat rare missing counts as a single visit
6: return  $E^+, E^0$ 

```

```

def log_marginal_over_lamgrid(y, lam_grid, p, n_trunc_extra=30):
    y = np.asarray(y)
    n_max = int(y.max()) + n_trunc_extra
    n_range = np.arange(int(y.max()), n_max + 1)          # (Nc,)

    # log P(y_k | N) summed over replicates k, independent of lambda
    log_lik_k = binom.logpmf(y[:, None], n_range[None, :], p) # (K, Nc)
    log_lik_sum = log_lik_k.sum(axis=0)                     # (Nc,)

    # log Poisson prior over N for every lambda in the grid at once
    log_prior_N = poisson.logpmf(n_range[:, None], lam_grid[None, :]) # (Nc, L)

    log_joint = log_lik_sum[:, None] + log_prior_N          # (Nc, L)
    m = log_joint.max(axis=0, keepdims=True)
    log_ml = (m + np.log(np.sum(np.exp(log_joint - m),
                                   axis=0, keepdims=True))).ravel()

    return log_ml # (L,)

```

Listing 1: Vectorized log marginal likelihood $\log P(y_g | \lambda, p)$, evaluated simultaneously for every λ on a grid via NumPy broadcasting.

2.4. Flowering-based confidence

Raw flower counts (NO_FLWS) are heavily right-skewed (median 2.4, maximum 1,560 across the flowering survey), meaning direct min–max normalization collapses most values toward zero. We instead compute

$$\text{confidence_flowering} = \min_{\text{max}}_{\text{within-sp.}} \left(\log(1 + \text{NO_FLWS}) \right), \quad (5)$$

normalizing within each plant species rather than globally, as absolute flower counts are not directly compara-

ble across species with divergent floral morphology. Algorithm 5 implements this transformation, including the degenerate case where a species exhibits zero variation in flower count across the survey.

2.5. Joining visitation and flowering records

Visitation and flowering records are joined on (PLOT_ID, YEAR, WATCH, PLTSP_CODE). For the small fraction of watch-level interaction records lacking a concurrent flowering survey (0.24% of the final event stream), we fall back to the most recent prior flowering observation for that species in that plot-year (last-observation-carried-forward within

Algorithm 3 BuildReplicateGroups — assemble the watch-level replicate count vector y_g for every (plot, year, plant, pollinator) group

Require: positive interactions E^+ , harmonized visitation table vis

Ensure: list of groups g , each with replicate vector y_g and replicate count K_g

```

1:  $watchTotals \leftarrow \text{GroupSum}(E^+, \text{keys} = [\text{plot}, \text{year}, \text{plant}, \text{poll}, \text{watch}], \text{value} = \text{NO\_INT})$ 
2:  $allWatches \leftarrow \text{GroupUnique}(vis, \text{keys} = [\text{plot}, \text{year}], \text{value} = \text{WATCH})$  ▷ full replicate set per plot-year
3: for all (plot, year, plant, poll) in  $\text{GroupKeys}(watchTotals)$  do
4:    $watches \leftarrow allWatches[\text{plot}, \text{year}]$ 
5:    $y_g \leftarrow []$ 
6:   for all  $w \in watches$  do
7:      $\text{append } \{\text{NO\_INT} : (\text{plot}, \text{year}, \text{plant}, \text{poll}, w) \in watchTotals\}$  to  $y_g$  ▷ 0 if unrecorded on  $w$ 
8:   end for
9:    $K_g \leftarrow |watches|$ 
10:   $\text{emit group } \{\text{plot}, \text{year}, \text{plant}, \text{poll}, y_g, K_g\}$ 
11: end for

```

Algorithm 4 FitDetectionProbability — pool informative groups to identify $\hat{p}$, then compute a posterior mean $\hat{\lambda}_g$ for every group

Require: groups from Algorithm 3; grids $p_1, \dots, p_{19} \in [0.05, 0.95]$ and $\lambda_1, \dots, \lambda_{150}$ (geometric, $[0.01, 80]$)

Ensure: pooled detection probability $\hat{p}$; posterior mean $\hat{\lambda}_g$ for every group

```

1: informative  $\leftarrow \{g : K_g \geq 2\}$ ; sample up to 250 groups from informative
2: for all  $p_i$  in the  $p$ -grid do
3:    $total_i \leftarrow 0$ 
4:   for all  $g$  in the sampled informative groups do
5:      $\ell \leftarrow \text{LogMarginalOverLamGrid}(y_g, \{\lambda_j\}, p_i)$  ▷ Listing 1, covers the whole  $\lambda$ -grid
6:      $total_i \leftarrow total_i + \max_j \ell_j$  ▷ profile out  $\lambda_g$ 
7:   end for
8: end for
9:  $\hat{p} \leftarrow \arg \max_i total_i$ 
10: for all  $g$  in all groups do
11:    $\ell \leftarrow \text{LogMarginalOverLamGrid}(y_g, \{\lambda_j\}, \hat{p})$ 
12:    $w_j \leftarrow \exp(\ell_j - \max_j \ell_j)$ ;  $w_j \leftarrow w_j / \sum_j w_j$  ▷ normalize to posterior over the  $\lambda$ -grid
13:    $\hat{\lambda}_g \leftarrow \sum_j w_j \lambda_j$  ▷ posterior mean
14: end for
15: return  $\hat{p}, \{\hat{\lambda}_g\}$ 

```

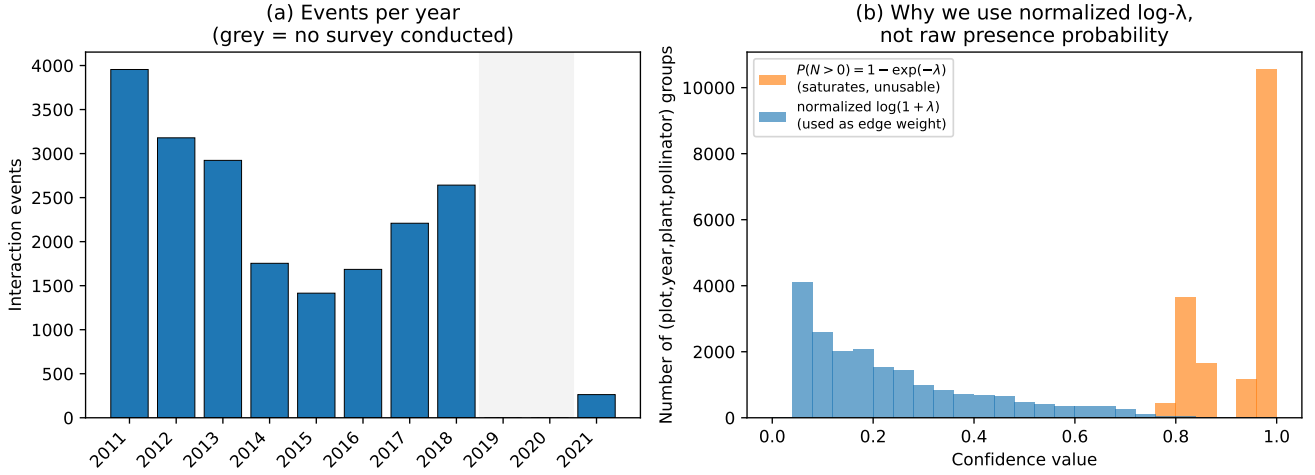


Figure 1: (a) Interaction events per year; grey bars mark years with no field survey (2019–2020), representing the sampling gap that breaks naive calendar-year splitting. (b) Distribution of the raw N -mixture presence probability $P(N > 0)$ (orange, saturates near 1 for almost all 17,472 groups) versus the normalized $\log(1 + \lambda)$ confidence used as the edge weight (blue, retains graded variation).

season), flagged via `is_imputed_flowering` for downstream sensitivity analysis. Records with no preceding observation receive a neutral confidence value of 0.5. Algorithm 6 outlines this join-with-fallback procedure.

2.6. Event stream assembly and temporal axis

Each retained interaction is emitted as a tuple with elapsed time τ , plant and pollinator identifiers, and confidence weight c , i.e., $(\text{tau}, \text{plant_id}, \text{pollinator_id}, c)$,

Algorithm 5 FloweringConfidence — log-transform and within-species normalize

Require: flowering table flw

Ensure: flw augmented with `confidence_flowering` $\in [0, 1]$

```

1:  $flw.NO\_FLWS \leftarrow \text{FillNA}(flw.NO\_FLWS, 0)$ 
2:  $flw.log\_flws \leftarrow \log(1 + flw.NO\_FLWS)$ 
3: for all species  $s$  in  $flw.PLTSP\_CODE$  do
4:    $S \leftarrow \{r \in flw : r.PLTSP\_CODE = s\}$ 
5:   if  $\max(S.log\_flws) - \min(S.log\_flws) < 10^{-9}$  then
6:      $S.confidence\_flowering \leftarrow 0.5$  ▷ degenerate case: no variation within species
7:   else
8:      $S.confidence\_flowering \leftarrow \frac{S.log\_flws - \min(S.log\_flws)}{\max(S.log\_flws) - \min(S.log\_flws)}$ 
9:   end if
10: end for
11: return  $flw$ 

```

Algorithm 6 JoinWithFallback — attach flowering confidence, with last-observation-carried-forward

Require: watch-level interactions *interactions*, flowering confidence table *flwConf* (Algorithm 5)**Ensure:** *interactions* augmented with `confidence_flowering` and `is_imputed_flowering`

```
1: merged  $\leftarrow$  LeftJoin(interactions, flwConf, on = [plot, year, watch, plant])
2: merged.is_imputed_flowering  $\leftarrow$  IsNull(merged.confidence_flowering)
3: for all r  $\in$  merged where r.is_imputed_flowering = true do
4:   cand  $\leftarrow$  {c  $\in$  flwConf : c.plot = r.plot, c.year = r.year, c.plant = r.plant, c.date  $\leq$  r.date}, sorted by date
5:   if cand  $\neq$   $\emptyset$  then
6:     r.confidence_flowering  $\leftarrow$  cand[last].confidence_flowering ▷ most recent prior record
7:   end if
8: end for
9: merged.confidence_flowering  $\leftarrow$  FillNA(merged.confidence_flowering, 0.5) ▷ no prior record exists
10: return merged
```

Algorithm 7 BuildEventStream — assemble the final weighted temporal event stream

Require: positive interactions E^+ , posterior confidences from Algorithm 4, flowering confidences from Algorithm 5**Ensure:** sorted event stream $\{(\tau, \text{plant_id}, \text{pollinator_id}, c)\}$

```
1: watchLevel  $\leftarrow$  GroupSum( $E^+$ , keys = [plot, year, watch, plant, poll, date], value = NO_INT)
2: watchLevel  $\leftarrow$  LeftJoin(watchLevel,  $\{\hat{\lambda}_g \rightarrow \text{confidence\_posterior}\}$ , on = [plot, year, plant, poll])
3: watchLevel  $\leftarrow$  JoinWithFallback(watchLevel, flwConf) ▷ Algorithm 6
4: watchLevel.c  $\leftarrow$  watchLevel.confidence_posterior  $\times$  watchLevel.confidence_flowering
5: t0  $\leftarrow$  min(watchLevel.date)
6: watchLevel. $\tau$   $\leftarrow$  (watchLevel.date - t0).days ▷ continuous axis, not the WATCH ordinal
7: return SortBy(watchLevel,  $\tau$ )
```

Algorithm 8 ChronologicalSplit — leakage-safe train/validation/test partition

Require: event stream *events* from Algorithm 7; fractions *trainFrac* = 0.8, *valFrac* = 0.1**Ensure:** *train*, *val*, *test*, non-overlapping and chronologically ordered

```
1: events  $\leftarrow$  SortBy(events,  $\tau$ )
2: n  $\leftarrow$  |events|
3: trainEnd  $\leftarrow$   $\lfloor n \times \text{trainFrac} \rfloor$ 
4: valEnd  $\leftarrow$   $\lfloor n \times (\text{trainFrac} + \text{valFrac}) \rfloor$ 
5: train  $\leftarrow$  events[0 : trainEnd]
6: val  $\leftarrow$  events[trainEnd : valEnd]
7: test  $\leftarrow$  events[valEnd : n]
8: return train, val, test
```

where τ represents the elapsed days since the first survey observation (2011-07-12) — rather than the WATCH ordinal, which would obscure variable inter-survey durations and multi-year gaps — and $c = \text{confidence_posterior} \times \text{confidence_flowering}$. Watch-level totals are obtained by summing NO_INT across duplicate within-watch minute-level records prior to this step. Algorithm 7 details the final stream assembly.

2.7. Chronological split

Due to the 2019–2020 survey gap, a fixed calendar-year partition boundary (e.g., train $\leq$ 2018, validation = 2019, test $\geq$ 2020) silently produces an empty validation split. We instead split by cumulative event volume along the continuous τ axis (80/10/10), as detailed in Algorithm 8, ensuring robustness against irregular sampling gaps. The resulting train, validation, and test partitions contain 16,085, 2,011, and 2,011 events, respectively, spanning 2011–2017, 2017–2018, and 2018/2021. Splitting mid-year at train/validation and validation/test boundaries introduces no data leakage, as the partition strictly respects elapsed time τ .

Data Availability

The processed dataset described in this article will be deposited upon manuscript acceptance in the Ecological Frontiers data community hosted by Science Data Bank (ScienceDB) at <https://test.scidb.cn/en/c/ef>, per the journal’s Data Paper submission requirements; the DOI will be added at that stage. The underlying raw data are publicly available from the Environmental Data Initiative repository as data package `knbn-1ter-and.5216`, released under a CC BY 4.0 license.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude for paraphrasing. The authors reviewed and edited the output as needed and take full responsibility for the content of the published article.

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